Kruti Shah

Colorado, USA 80304
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Education

University of Colorado Boulder Boulder, CO
Master of Science in Computer Science, GPA: 3.767/4.0 Aug 2025 – May 2027

Dwarkadas J. Sanghvi College of Engineering Mumbai, India
B.Tech in Computer Engineering with Honors in Intelligent Computing, GPA: 9.05/10.0 Aug 2021 – May 2025

Skills

Languages/Frameworks: Python, Java, C++, C, JavaScript, TypeScript, FastAPI, Flask, Django, React.js, Node.js, REST APIs
Machine Learning: TensorFlow, PyTorch, scikit-learn, OpenCV, NLP, Transformers, RAG, LangGraph
Databases/Tools: MySQL, PostgreSQL, MongoDB, Git, Docker, Terraform, Pytest, AWS, GCP, Linux, Kubernetes, CI/CD
Coursework: Data Structures and Algorithms, Software Engineering, Operating Systems, Computer Networks, Processor Organization and Architecture

Experience

Agentic AI intern May 2026 – July 2026
eClerx Services Limited, Remote

- **Shipped** an **8 node agentic AI system** using **LangGraph, Claude, and RAG** to automate financial analysis for a client spanning **14 companies**, combining structured extraction, semantic retrieval, and LLM reasoning to identify **financial trigger years**.
- **Designed** an end to end analysis workflow across multilingual document parsing, metric extraction, peer benchmarking, ratio analysis, ROE decomposition, and evidence backed risk assessment, turning annual reports into **automated investment reports**.
- **Iterated with client stakeholders**, translating requirements and feedback into extraction, retrieval, and reasoning improvements.

AI Intern Aug 2023 – Sep 2023
Assessli, Remote

- **Automated** student response evaluation by building a pipeline that classified submissions against reference solutions and generated evaluation reports, reducing manual grading effort by **40%** across **40+ partner schools**.
- **Optimized** a **Deep Reinforcement Learning recommendation** pipeline for personalized learning activities using longitudinal student data, improving recommendation performance by **18%**.

Python Intern Jul 2023 – Sep 2023
Emkay Global Financial Services Limited, Mumbai, India

- **Developed** a Python alerting system to flag **order discrepancies** within **15 seconds**, reducing manual monitoring by **30%**.
- **Automated** compliance analytics by parsing compressed transaction logs, computing order to trade ratios, and generating email reports, reducing analysis time by **50%**.
- **Trained** an LSTM on Infosys price data, identifying **5–7% pre-earnings** gains and **2–3% post-earnings** corrections.

Projects

AgentOS: Resilient Runtime for Autonomous LLM Agents August 2026 -Present

- **Engineered** a fault-tolerant agent runtime around **Claude** with a **ReAct loop** and native tool calling, orchestrating **16 tools across 8 MCP servers** spanning filesystem, database, web, shell, Git, HTTP, and Python execution.
- **Built** a self-healing recovery pipeline using **backoff retries, argument repair, tool fallbacks, query expansion, and loop detection**, improving the **benchmark task completion rate from ~30% to 100% across 100 fault-injection scenarios**.
- **Implemented** secure agent execution with **policy based access control, prompt injection isolation, filesystem sandboxing, and human approval gates**, backed by FastAPI, React, SSE, DuckDB, Docker, and pytest.

ChronoBase: Git for Databases (Cloud Native Data Versioning Platform) February 2026 – April 2026

- **Designed and built** a Git like database system supporting **snapshots, branching, and point in time restoration** of the states.
- **Automated AWS infrastructure** with **Terraform**, using S3, SQS, EventBridge, and DocumentDB to orchestrate asynchronous snapshot processing and persistent storage.
- **Provisioned** backend services on **EC2** with encrypted and compressed MongoDB ingestion and multi tenant isolation, and instrumented **Grafana** dashboards for resource usage, errors, logs, and performance.

BoulderMove: Smart Multimodal Trip Planning System November 2025 – December 2025

- **Built** a cloud native trip planning system combining **GTFS/RAPTOR** transit and **OSMnx** pedestrian routing with **Pytest** coverage.
- **Developed** a decoupled **XGBoost** inference service on **Google Cloud Run**, achieving **87% accuracy** in predicting transit on time arrivals using route, schedule, and weather, event data from **OpenWeatherMap and Ticketmaster APIs**.
- **Deployed** the React frontend to **Netlify** and containerized backend services on **Render and Google Cloud Run**, using **Cloud Build, Artifact Registry, Compute Engine, Cloud SQL.**

Publications & Patents

- **Explainable AI for Yoga Pose Detection and Correction**, *Educational Administration: Theory and Practice*, 2024.
- **EcoBot: Pest Detection Alert System**, book chapter, *Technologies for Energy, Agriculture, and Healthcare*, 2026.
- **HazardScout:** Utility patent published (202521045231) and design patent filed (458534-001), **gas leak detection robot**.

Achievements & Co-Curricular Activities

- Advanced to the **pre-finale (Top 70 of 29,000+ teams)** in the **National Myntra HackerRamp WeForShe 2024 hackathon**.